

Quote Tool — v1 Companion Guide

A first proof-of-concept — one unified quote tool with a smart item-code finder built in. What it does, what to trust, and what's next.

The problem we're attacking. Today, to quote a profiled (non-lab-pack) waste stream, Kristin scrolls ~760 item codes by hand to find the right one. This tool narrows that to a short list in seconds — ranked by what RedArc has *actually billed* for similar waste, with the cheaper facility options and the reasoning shown. **This is a v1: built to prove the approach is real, not to quote live yet.**

How it works

We joined Zak's five data files into one catalog (761 codes + pricing + routing). The key ingredient is **889 real waste-stream profiles** — each already labeled with the item code RedArc billed and the facility it went to. That's an answer key. So instead of hand-coding every routing rule, the tool:

- Reads your waste description (a manifest shipping line works great — UN number, hazard class, packing group).
- Finds the most similar *real past profiles* and recommends the code *they* were billed under.
- Shows the facility options ranked by cost, plus Zak's routing notes (preferred / backup facility).
- Gives a **confidence flag** — and when it's unsure, it says "confirm with Zak" instead of guessing.

How accurate is it? (measured, not claimed)

We replayed all 889 real profiles through the tool (hiding each one's answer) and checked whether it recommended the code that was actually billed:

72%

right code in the top-3 shortlist

74%

top-1 correct on high-confidence calls

889

real profiles as the answer key

Why top-3 is the number that matters: the job is to narrow 760 codes to a shortlist. Handing Kristin a 3-code shortlist that contains the right answer 72% of the time — from a waste description alone — already replaces most of the scrolling. The confident cases (~40%) can auto-fill; the rest still get a short list instead of the whole catalog. **This is a baseline to beat, and every future improvement is measurable against it.**

Using the tool

It's now **one unified quote tool**, not two pages. You start a quote (client, A/B/C tier, service, generator, carrier), then click **Add waste** — the finder opens right there. Paste a waste description (or click an example) and you'll see the recommended item code, a confidence badge, the facility options ranked by cost, Zak's routing notes, and the real waste streams it matched on. Pick a facility and it drops a priced disposal line; add transport, labor and supplies and the totals update live at your tier. Try a manifest shipping line like **UN1993, Waste Flammable liquids n.o.s. (acetone), 3, PGII**.

Two ways to open it: a **secured link** (RedArc email sign-in) that runs on real prices and your real accounts, or a **public preview** (no login) that's de-identified with relative price bands. Both are linked from the project dashboard.

Caveats — please read before judging

It's v1 on raw data. Item-code descriptions are inconsistent (generated from vendor rate sheets, never cleaned). We did *not* clean them first — cleanup is a later step. The number above is despite the mess, not because the data is tidy.

Two versions, by design. The **public preview** is de-identified — no customer names, prices as relative bands (\$ = cheapest) — because the public dashboard isn't a place for RedArc's pricing. The **secured link** (behind a RedArc email sign-in) runs the real per-container A/B/C prices and real account names. Real data never touches the public site.

The hardest layer isn't in yet. No SDS/chemistry reasoning, cost-optimization, or facility acceptance-criteria (e.g. Tulsa's 500 ppm VOC limit) — and the approvals-department conservatism that lives only in Zak's head is deliberately left as a "low-confidence → confirm" flag, not a guess.

Not for production quoting. This proves direction. It is not yet a tool Kristin should quote a customer from unaided.

A few questions for Zak

- 1. Does the shape fit?** Is "short list of 3 + a confidence flag, auto-fill when sure, confirm-with-Zak when not" the right way for Kristin to work?
- 2. Two routing corrections to confirm** (from your Operational Color notes): the HF→Tulsa code is still in the recycling group (remove?), and recycled petroleum shouldn't be under an FBL code to VAL — master-list error or a rogue profile?
- 3. Where should it live?** A Claude skill Kristin invokes (like the lab-pack sorter), or a small internal web tool? Either is easy.
- 4. Does the end-to-end flow fit?** Now that it's on a secured link with real prices and real accounts, does building a quote this way (header → add waste → transport, labor & supplies → live totals) match how you'd actually put one together?

What's next

The accuracy levers we haven't pulled yet, each measurable against the 72%: (1) match on hazard class + RCRA/state codes more deeply, (2) Claude's chemistry/SDS reasoning, (3) the cost-optimization and acceptance-criteria routing layers Zak described. Then: wire in the lab-pack skill so one flow handles both lab packs and profiled waste end-to-end.